URDU DUB STUDIO

Project File Map

How the source files handle a dub and how Docker starts the app.

01 / WHEN YOU CLICK START

index.html — Browser interface

Accepts the YouTube link, sends it to Flask, and polls for status.

↓

app.py — Flask backend

Validates the link, starts the job, and serves status and downloads.

↓

dubbing.py — Dubbing pipeline

Imports the video, transcribes English, and renders the Urdu dub.

↓

local_translation.py

Gets Urdu script and Roman Urdu via Ollama.

→

Speech and video tools

Piper makes the voice. FFmpeg joins it to video.

OUTPUT Video with Pakistani Urdu audio (MP4) + Roman Urdu transcript (TXT)

02 / WHEN DOCKER STARTS

compose.yaml

Defines the services, startup checks, ports, and storage.

ollama

Starts the local model server.

→

model-setup

Waits for Ollama, pulls Qwen, and exits successfully.

→

app

Starts Flask after model setup succeeds.

BUILD FILES

Dockerfile builds the app image. requirements.txt lists Python packages.